

Group TYG19: Individual Work Distribution

1. Harsh Manjramkar (Team Lead & Core Architect)

Primary Folders Owned: `src/gateway/` , `src/hitl/` , and `static/`

Harsh handled the core infrastructure, backend architecture, and full-stack integration.

- Core Proxy Architecture: Designed and implemented the main Transparent Proxy gateway using FastAPI (`src/gateway/router.py`) to flawlessly intercept JSON-RPC traffic.
- Asynchronous HITL Manager: Engineered the complex Human-in-the-Loop workflow (`src/hitl/manager.py`) using `asyncio.Event()` to securely pause and resume execution in server memory.
- Frontend Logic & AI Integration: Developed the JavaScript architecture for the Visual E2E Flow Simulator and the Live Custom Terminal (`static/index.html`), and handled the live integrations with Claude Desktop, Cursor IDE, and Antigravity.

2. Vedant Gaidhani (Security Engineer - Threat Detection)

Primary Folders Owned: `src/risk/`

Vedant owned the inbound payload inspection and prompt injection defense mechanisms.

- Regex Detection Engine: Built the comprehensive regex signature database (`src/risk/detectors.py`) to instantly identify and block prompt injections, DAN jailbreaks, and SQL injection vectors.
- Dynamic Heuristics: Implemented the wildcard heuristic engine (`src/risk/heuristics.py`) to dynamically flag dangerous tool executions (like 'delete', 'drop', or 'execute').
- Risk Router: Co-developed the `evaluator.py` logic to accurately assign LOW, MEDIUM, or HIGH risk severity to incoming AI payloads.

3. Vineet Wagh (Security Engineer - Data Privacy)

Primary Folders Owned: `src/sanitizer/` and `static/`

Vineet owned the outbound data inspection, sanitization, and UI/UX design.

- PII Sanitization Engine: Developed the recursive JSON parsing algorithm (`src/sanitizer/pii_engine.py`) that actively intercepts returning payloads from the MCP server.
- Redaction Logic: Created the regex patterns (`src/sanitizer/patterns.py`) to detect and scrub sensitive data like Aadhaar numbers, PAN cards, and API keys without corrupting the JSON schema.
- Dashboard UI/UX: Wrote the advanced CSS styling and animations (`static/index.html`) for the interactive dashboard and data-packet simulation.

4. Arjun Joshi (QA & Integration Testing Engineer)

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Primary Folders Owned: tests/ and tests/integration/

Arjun owned the massive testing suite to ensure the proxy was enterprise-ready.

- Enterprise Target Mocking: Designed the Mock Target Hub that correctly simulates the exact JSON-RPC schemas of 15 Enterprise MCPs (GitHub, PostgreSQL, Slack, Docker, etc.).
- Automated Test Suite: Wrote and maintained all 57 automated pytest integration tests (tests/integration/test_15_real_mcp_servers.py), proving the proxy never corrupts safe payloads.
- Documentation & DevOps: Handled the generation of the PDF demonstration guides, project documentation, and repository structure cleanup.